

Xuankun Yang

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EDUCATION

Shanghai Jiao Tong University
BEng in Artificial Intelligence

Shanghai, China
Sep 2023 – Present

- CGPA: 4.01 / 4.3, Rank 5 / 98
- Average score: 92.01 / 100, Rank 7 / 98

RESEARCH INTERESTS

- Foundation models
- Mechanistic interpretability
- Efficient model training and inference

EXPERIENCE

SJTU Artificial Intelligence Institute DeepVision Lab
Research Intern, PEFT

Shanghai, China
Oct 2024 – May 2025

- Mentor: [Wei Shen](#)
- Co-authored *Weight Spectra Induced Efficient Model Adaptation* (under review), a PEFT method motivated by empirical analysis of fine-tuning dynamics that modulates top singular directions for more effective adaptation.
- Validated the method across language, commonsense reasoning, and vision benchmarks including GLUE, Commonsense170K, and VTAB-1K, achieving strong performance under strict parameter budgets.

SJTU Artificial Intelligence Institute DeepVision Lab
Research Intern, Model Merging

Shanghai, China
Jul 2025 – Present

- Mentor: [Wei Shen](#)
- Conducted systematic research on model merging through extensive literature review and empirical investigation of existing paradigms, wrote a [technical survey](#).
- Authored *TIC: Pairwise Task Interference Calibration for Model Merging* (under review), localizes directional interference and consistently improved most merging baselines across vision and language tasks.

SJTU Artificial Intelligence Institute DeepVision Lab
Research Intern, Visual Token Pruning

Shanghai, China
Oct 2025 – Jun 2026

- Mentor: [Wei Shen](#)
- Co-first authored *Entropy-Aware Dense Visual Token Pruning* (ECCV 2026), proposing dense semantic relevance modeling, entropy-assisted denoising, and submodular token selection with spatial priors to preserve fine-grained cues under strict token budgets.
- Demonstrated state-of-the-art accuracy-efficiency trade-offs across LLaVA, LLaVA-NeXT, LLaVA-Video, and Qwen-VL families, with strong latency and FLOPs reduction under aggressive pruning ratios.

SJTU School of Artificial Intelligence M³ Lab
Research Intern, World Action Models

Shanghai, China
Jun 2026 – Present

- Mentor: [Yanwei Li](#)
- Exploring World Action Models (WAMs) for embodied intelligence, with a focus on action-conditioned video prediction, environment dynamics modeling, and their potential applications to planning and decision-making.

PAPERS

Weight Spectra Induced Efficient Model Adaptation

Second Author, Under review.

May 2025

Combating Textual Noise and Redundancy: Entropy-Aware Dense Visual Token Pruning

Co-first Author, ECCV 2026.

Mar 2026

TIC: Pairwise Task Interference Calibration for Model Merging

First Author, Under review.

June 2026

PROJECTS

A RAG Teaching Assistant System

Jan 2026

- Built a multimodal RAG TA system with multimodal embeddings, cross-modal retrieval and source-grounded citation to answer questions over course materials containing text, images, charts, and formulas.
- Constructed a small evaluation set and quantified both retrieval and generation quality.

Data-Efficient Vision Transformer on CIFAR-10

Jun 2025

- Studied Vision Transformer on the small-sample CIFAR-10 setting, focusing on data-efficient learning; implemented native ViT and ResNet+ViT hybrid variants with configurable patch/head settings and stochastic depth.
- Reproduced and extended ViT-paper-style analyses on attention maps, attention distance, gradient rollout, embedding similarity, etc., and documented the findings in a [technical blog](#).

End-to-End Voice Activity Detection Pipeline

May 2025

- Built an end-to-end voice activity detection pipeline covering pre-emphasis, framing, windowing, acoustic feature extraction, frame-level classification, and time-domain speech segment restoration.
- Conducted extensive explorations over frame length/shift, preprocessing, smoothing, and model choices, and summarized the analysis in a [technical blog](#).

SKILLS

Programming: Python, C++

Deep Learning: PyTorch, Hugging Face

Tools: Linux, Git, L^AT_EX, TensorBoard

LLM/VLM Frameworks: LLaVA, Qwen-VL, lmms-eval